Title: Your Old Things Is Someone’s new Treasure

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(DEEMED TO BE UNIVERSITY)

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Abstract

- Web-based trading platform for exchanging unused everyday items
- No money involved - pure value exchange platform
- Focus on sustainability and reducing waste
- Builds community connections through item trading
- Two components: Frontend (HTML/CSS/JS) and Backend (Java DSA)
- Frontend features glass morphism UI with LocalStorage persistence
- Backend implements all core data structures and algorithms
- Hash Table for O(1) user authentication
- Linked Lists for flexible item storage
- Stack for search history management
- Queue for trade request processing

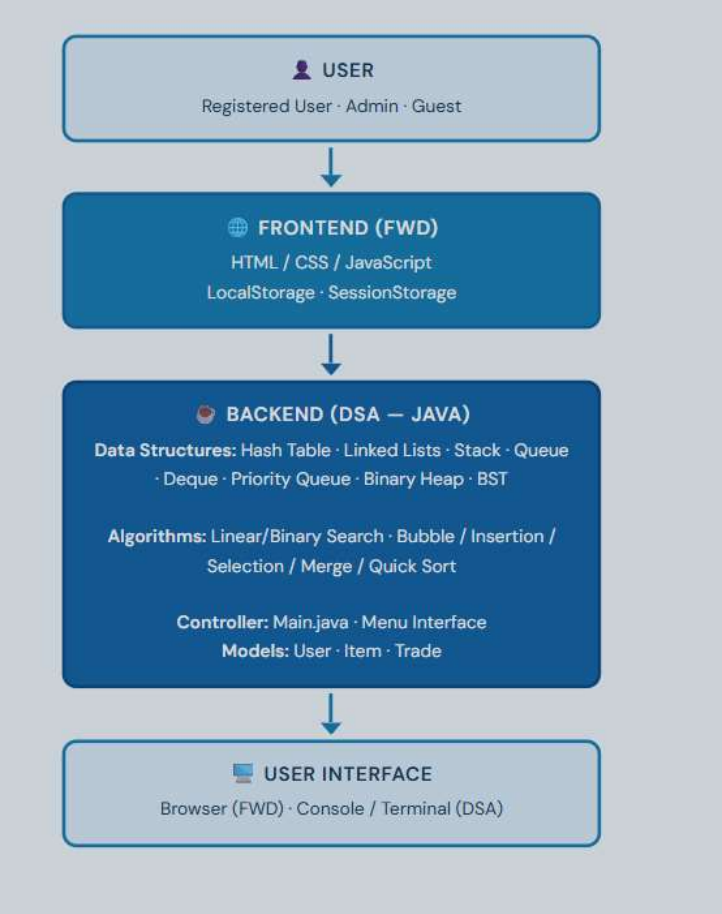
Existing System

- E-commerce platforms focus on monetary transactions
- Social media marketplaces lack structured data
- Donation platforms offer one-way giving only
- No dedicated platform for item exchange without money
- Existing systems don't emphasize sustainability
- Limited community building features
- Items often discarded instead of reused
- No focus on circular economy principles
- Lack of data structure implementation for efficiency
- No time complexity analysis in current solutions

Proposed System

- Pure trading platform with no money involvement
- User authentication with login and signup functionality
- Item browsing with category filters (Kitchen, Tech, Clothing, Books, Tools)
- Real-time search and sort features
- Wishlist and cart management system
- Trade proposal and tracking functionality
- User profile dashboard with activity history
- Glass morphism UI with smooth animations
- Fully responsive design for all devices
- DSA backend with complete data structure implementations
- Time complexity analysis for all operations

System Architecture



Project Modules

- FWD
- Module 1: User Authentication - Login and signup forms with validation
- Module 2: Item Display - Grid layout with category filters
- Module 3: Search and Sort - Real-time filtering and sorting options
- Module 4: Wishlist Management - Add and remove favorite items
- Module 5: Cart Management - Add, remove, and total calculation
- Module 6: Trade System - Proposal and tracking functionality
- Module 7: User Profile - Dashboard with activity history
- Module 8: Data Persistence - LocalStorage for user data

DSA

Module 1: Linked Lists - Singly, Doubly, and Circular implementations

Module 2: Stack and Queue - LIFO and FIFO data structures

Module 3: Deque and Priority Queue - Double-ended and priority-based structures

Module 4: Binary Heap - Min-heap for price filtering

Module 5: Hash Table - Separate chaining for user authentication

Module 6: Binary Search Tree - Category organization

Module 7: Searching Algorithms - Linear and Binary search

Module 8: Sorting Algorithms - Bubble, Insertion, Selection, Merge, Quick sort

Objectives

- Create a sustainable trading platform with zero monetary exchange
- Implement beautiful, responsive frontend with glass morphism UI
- Demonstrate all five COs of Web Development course
- Implement all core data structures in Java from scratch
- Show time complexity analysis for each operation
- Provide real-world applications for each data structure
- Build community connections through item trading
- Promote environmental sustainability through reuse
- Reduce waste by giving old items new life
- Create comprehensive documentation for both components

Co’s Attainments - FWD

CO1: Internet Fundamentals, HTML & CSS
Client-server architecture, browser rendering, HTTP/HTTPS
Complete HTML5 structure with semantic tags (header, nav, section, footer)
CSS syntax, selectors, typography, colors, box model
CSS variables for consistent theming

CO2: Forms, Semantic Tags & Advanced CSS
Login/signup forms with validation
Flexbox for navigation, Grid for item display
Media queries for responsive design
CSS animations, transitions, glass morphism effects

CO3: JavaScript Programming Essentials
Ternary operators, if-else, switch statements
Loops: for, forEach
Functions: regular, arrow, callbacks
Arrays, objects, map, filter, find methods

CO4: JavaScript Interactivity, DOM & Events
DOM selection and manipulation
Event handling (click, input, submit)
LocalStorage for data persistence
JSON parsing and stringifying- converts the object to string for storage

CO5: Advanced Web Development & Deployment
Try-catch exception handling
Form validation with regex
SEO meta tags implementation
GitHub repository and version control

Co’s Attainments - DSA

CO1: Algorithm Analysis & Searching/Sorting
Linear Search O(n), Binary Search O(log n)
Bubble, Insertion, Selection Sort $O(n^2)$
Merge, Quick Sort O(n log n)
Empirical time measurement using System.nanoTime

CO2: Abstract Data Types with Linked Lists
Singly, Doubly, and Circular Linked Lists
Operations: append, prepend, insert, delete, search
Bidirectional traversal and wrap-around functionality

CO3: Stacks, Queues, Heaps & Priority Queues
Stack: push, pop, peek O(1) for search history
Queue: enqueue, dequeue O(1) for trade requests
Circular Queue for recently viewed items
Deque: addFront, addRear, removeFront, removeRear O(1)
Priority Queue and Binary Heap O(log n)

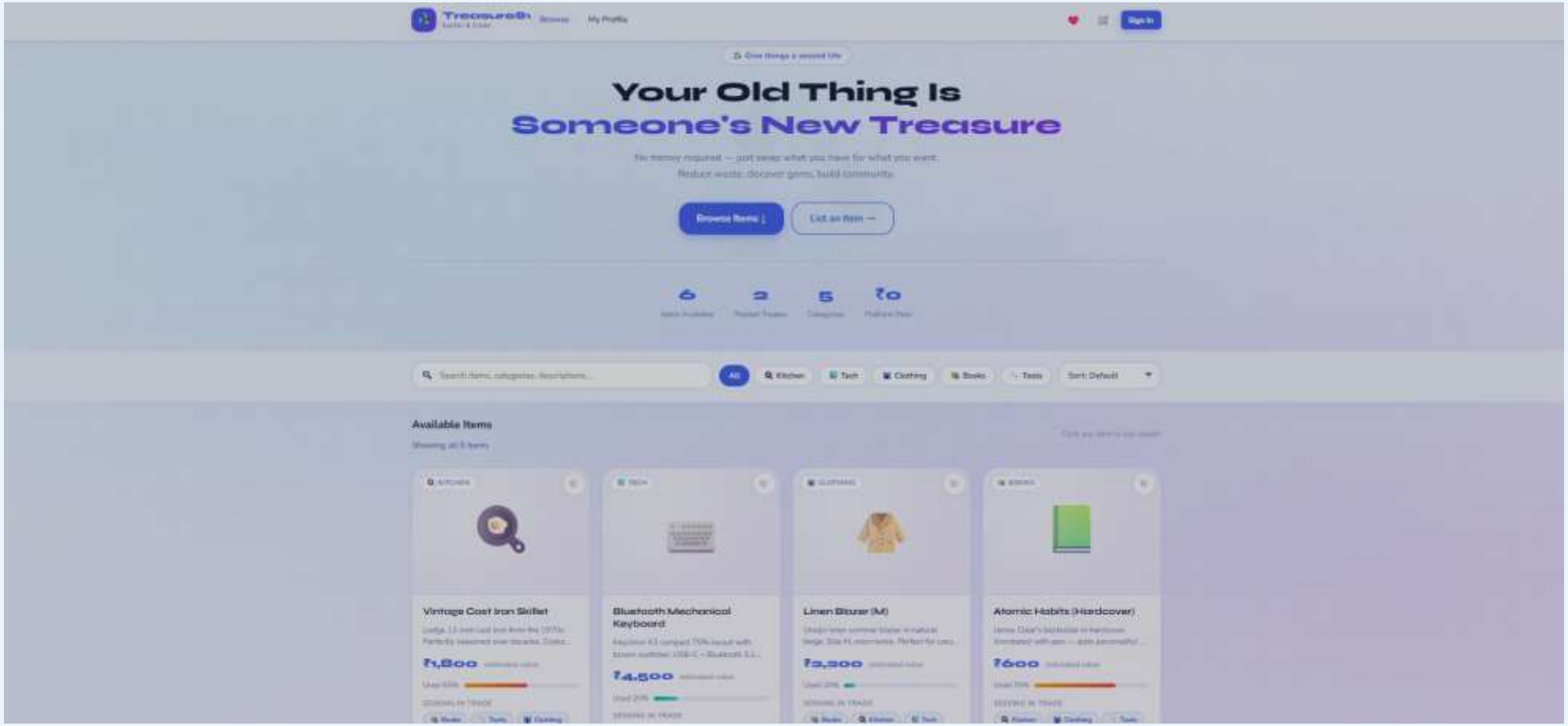
Co’s Attainments - DSA

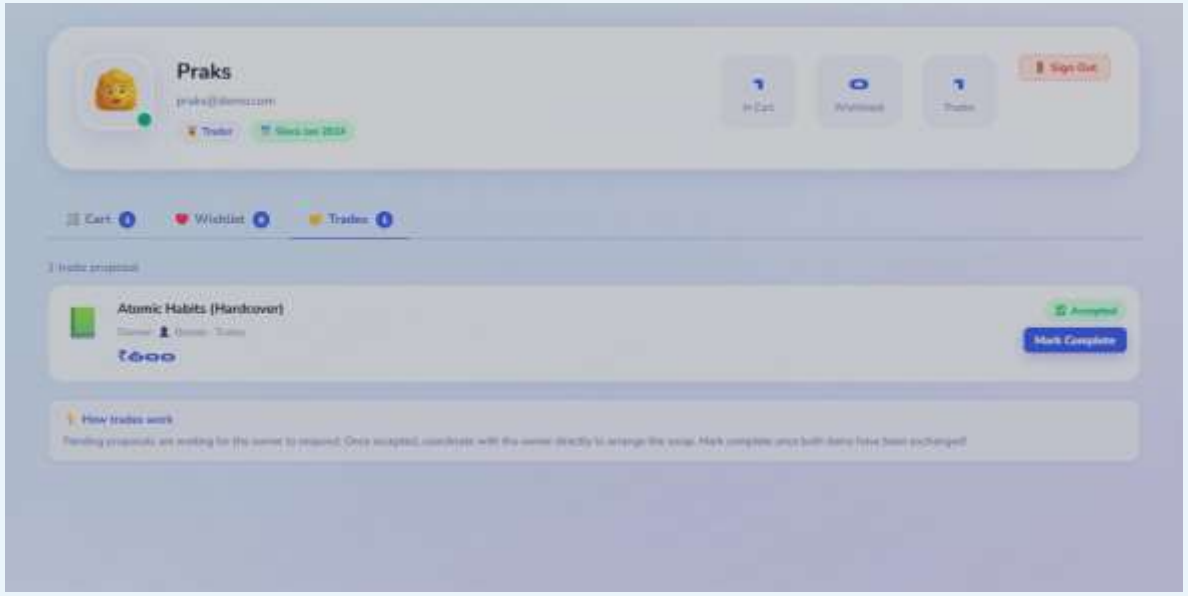
CO4: Hash-Based Data Structures
Hash Table with separate chaining
O(1) average for put, get, remove operations
Collision resolution using buckets
Dynamic resizing capability
Used for user authentication system

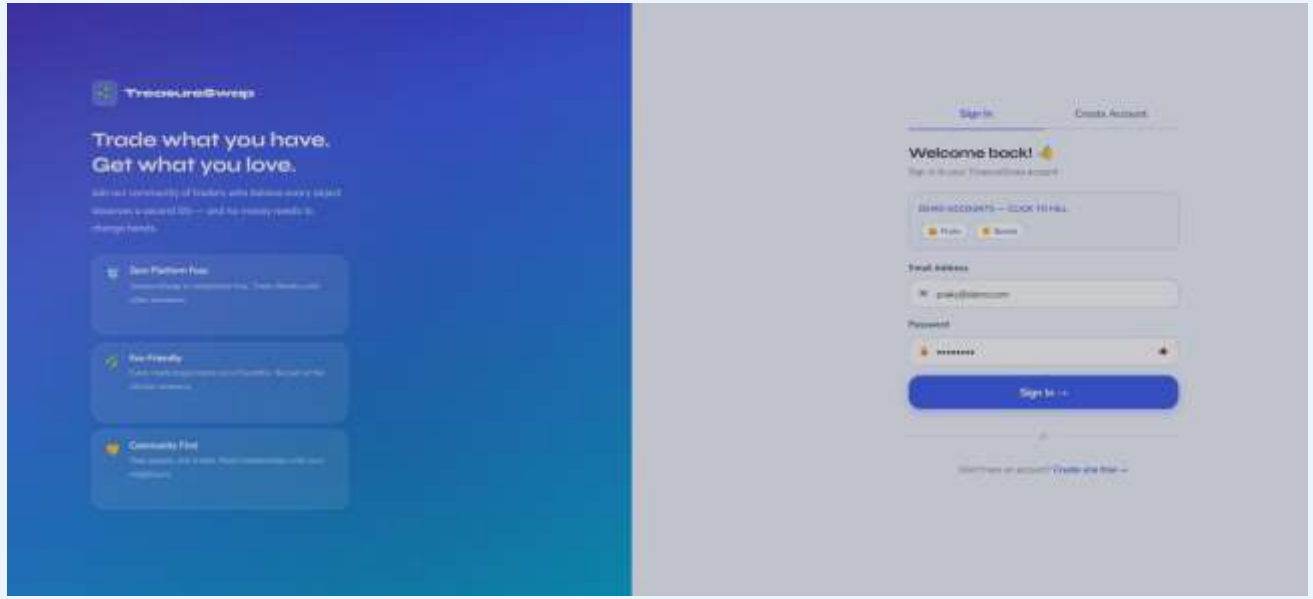
CO5: Practical Applications of Linear Data Structures
Stack: Search history (browser back button)
Queue: Trade requests (customer service line)
Circular Queue: Recently viewed (limited history)
Deque: Wishlist (priority shopping list)
Hash Table: User login (dictionary lookup)
Linked Lists: Item storage (playlist navigation)

CO6: Application Development Skills
Problem identification for trading platform
Requirements analysis for all features
Data structure selection based on use cases
Algorithm design with complexity analysis
Java OOP programming with generics
Exception handling and input validation
Menu-driven user interface
Complete working application with all structures

Screenshots of FWD




Praks
Cart
Wishlist
Trades
Atomic Habits (Hardcover)
Mark Complete
How trades work


Trade what you have.
Get what you love.
Welcome back!
Sign In

Screenshots of DSA

```
==================================================
  YOUR OLD THING IS SOMEONE'S NEW TREASURE
        DSA Implementation - Backend
==================================================

? Data Structures Initialized
   ? Hash Table: 4 users
   ? Doubly Linked List: 8 items
   ? Binary Search Tree: 6 categories


================================================
                MAIN MENU
================================================
1. ? Customer Login
2. ? Admin Login
3. ? New User Registration
4. ? Browse All Items (Linked List)
5. ? Search Items
6. ?? Filter by Category (BST)
7. ?? Wishlist (Deque)
8. ? Recently Viewed (Circular Queue)
9. ? DSA Overview
0. ? Exit
================================================
Enter choice:
```

```
================================================
Enter choice: 3

================================================
          NEW USER REGISTRATION
================================================
Username: prakshitha
Password: 123456
Full Name: Prakshitha Malla
City: hyd

? Registration successful! [Hash Table Insert: O(1

================================================
                MAIN MENU
================================================
1. ? Customer Login
2. ? Admin Login
3. ? New User Registration
4. ? Browse All Items (Linked List)
5. ? Search Items
6. ?? Filter by Category (BST)
7. ?? Wishlist (Deque)
8. ? Recently Viewed (Circular Queue)
9. ? DSA Overview
0. ? Exit
================================================
Enter choice:
```

Screehshots of DSA

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═══════════════════════════════════════════
              CUSTOMER LOGIN
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Username: prakshitha
Password: 123456

? Login successful! Welcome, Prakshitha Malla!
   [Hash Table Lookup: O(1) average - 28600 ns]

═══════════════════════════════════════════
     USER DASHBOARD - Prakshitha Malla
═══════════════════════════════════════════
1. ? Browse All Items
2. ? Search Items
3. ?? Filter by Category
4. ? Add New Item
5. ? View My Listings
6. ? View Trade Requests
7. ?? Wishlist (0 items)
8. ? Recently Viewed (0)
9. ? Logout
═══════════════════════════════════════════
Enter choice:
```

```
Enter choice: 1

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            ALL AVAILABLE ITEMS
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1. French Press Coffee Maker [kitchen] - ?1200 (Good)
   Stainless steel, 32oz capacity, used for 2 years
   Owner: user1

2. Wireless Earbuds [electronics] - ?1500 (Good)
   Good condition, 5hr battery life, with charging case
   Owner: user1

3. Unused Yoga Mat [sports] - ?500 (Mint)
   Brand new, non-slip, 6mm thickness, includes strap
   Owner: user2

4. Desk Lamp with USB [electronics] - ?800 (Good)
   Adjustable arm, warm/cool light settings, 3 USB ports
   Owner: user2

5. Cast Iron Skillet [kitchen] - ?900 (Good)
   Well-seasoned, 10-inch, perfect for cooking
   Owner: user3

6. Graphic Novel Collection [books] - ?1200 (Like New)
   12 volumes, various series, like new condition
   Owner: user3

7. Denim Jacket [clothing] - ?1500 (Good)
   Size M, dark wash, lightly worn
   Owner: user1

8. Basic Tool Set [tools] - ?2000 (Good)
   Hammer, screwdrivers, pliers, tape measure
   Owner: user2

   [Doubly Linked List Traversal: O(n) - 37708900 ns]

Enter item number to view details (0 to go back):
```

Conclusion

- FWD component successfully developed with glass morphism UI
- Complete authentication system implemented
- Item browsing, wishlist, cart, and trade features working
- Fully responsive design achieved
- All five FWD COs demonstrated effectively
- DSA component implements all core data structures
- Hash Table provides O(1) user authentication
- Linked Lists enable flexible item storage
- Stack manages search history efficiently
- Queue processes trade requests in order
- Binary Search Tree organizes categories
- All sorting and searching algorithms implemented
- Time complexity analysis for each operation
- Project promotes sustainability through reuse
- Technology fosters community connections
- Different data structures serve specific needs
- Time complexity understanding is crucial

Geo- Tag Photos

THANK YOU